

CHAPTER 2

PHYSICAL TRAINING

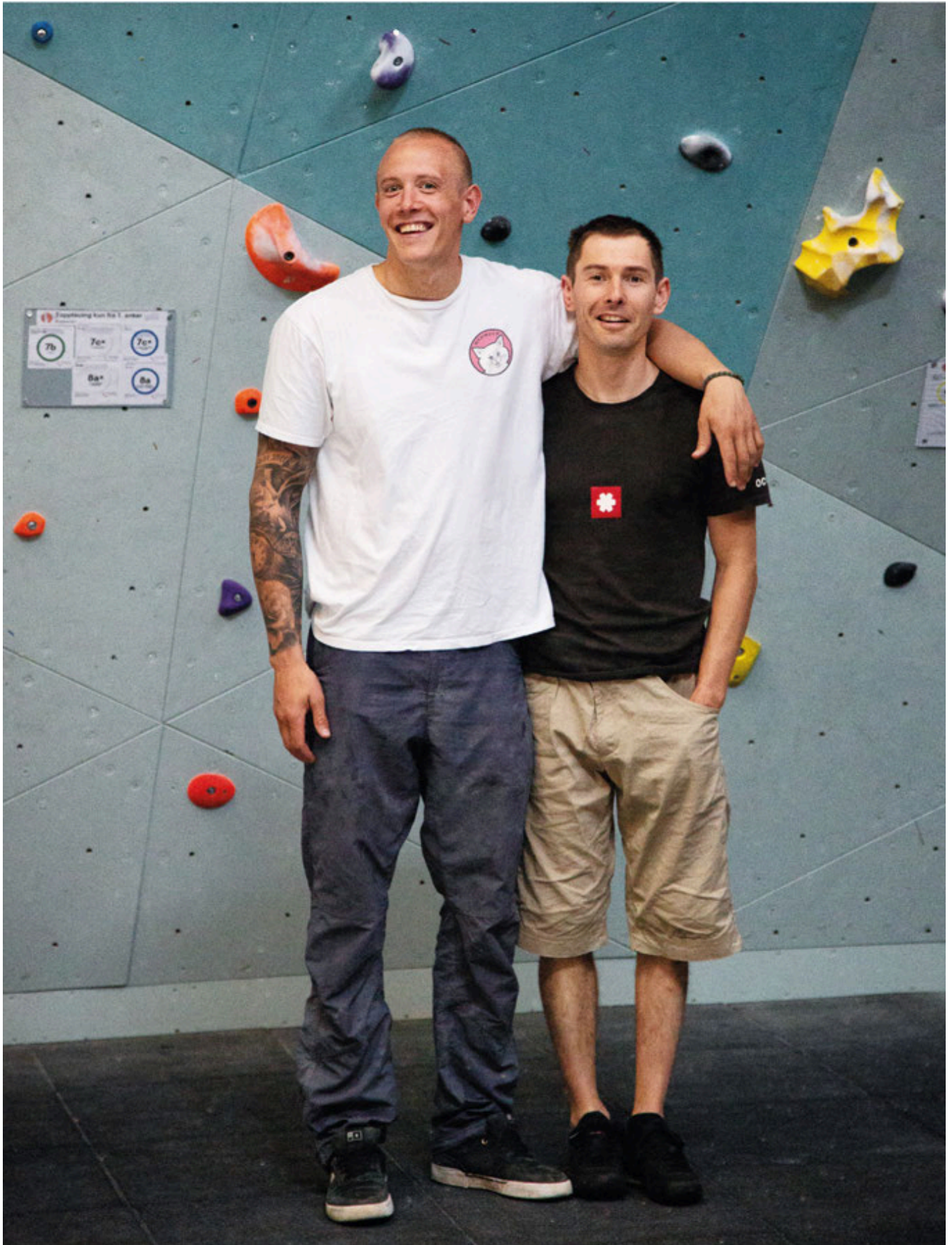


PHOTO: ALEX MANELIS

Martin Moberåten on Spectre (Font 8b), Bishop, USA.

OUT OF ALL the different factors influencing a climbing achievement, the physical aspect is perhaps the most talked about. And not without reason. As climbing becomes progressively harder, the holds get worse and further apart. Walls get steeper and the cruxes more sustained. This places increasing demands on physical factors like finger strength, arm strength, core strength, endurance and mobility. These are factors that are relatively easy to train, and progression can be just as easily measured. If we look at the traditional approach of splitting climbing performance into three, the physical factors are considered to be more important for performance than the technical and mental factors.

This chapter starts by looking at what defines ‘the climbing body’. Next up is the different physical performance factors within strength, endurance and mobility, and how these can be trained.



Kenneth Elvegård and Adam Pustelnik are both skilled climbers with different physical attributes.

THE CLIMBING BODY

Every climber has different physical proportions and attributes, yet to this day these seem not to make much difference to a climber's performance. Even at the elite level, there are huge differences in height, arm length and body weight, evidence that trainable traits are more important to performance than how a climber's body is put together anatomically. Still, climbing does favour a relatively slender body type with a particular distribution of muscle mass – the bulk of the muscle in the upper body rather than the legs. This is why climbing can be seen as a weight-dependent sport, where a lower body weight will allow a climber to hold on to smaller holds since they have fewer kilos to haul.

It can therefore be tempting to lose weight in order to improve performance, rather than exercising to improve the correlation between strength and body weight. This weight loss has traditionally been common in climbing. Nevertheless, we have in recent years seen a shift towards steeper climbing, bouldering and an increased focus on technically complex and demanding climbing. We don't doubt that losing weight is performance enhancing, but shedding pounds can also have a negative effect, as less muscle mass will make it harder for you to do physically hard moves on steep terrain. Also, it won't be of any help to be thin and lightweight if you're not technically adept enough to do tricky moves, and you'll probably be just as scared when you're 10 feet above the last bolt no matter how much you weigh! For adult climbers, a reduction in weight can be an important performance variable, but we recommend you focus more on developing your technical, tactical, physical and mental abilities before you consider losing weight to boost your performance. Losing weight also means you'll be training with an energy deficit for a period of time, and this will make it difficult to lay down the right amount of quality training you need, which is what you should be focusing on. The effect of a reduction in weight will also be much higher if your skills are already at a high level, so instead, consider losing weight only as a last tweak of form before a project or competition, and only after you're at a stable and acceptable performance level. In addition, for younger athletes, there are some obvious health risks associated with losing weight, and on a general basis, we recommend that young climbers don't lose weight in order to increase performance. We'll cover this in greater detail in Chapter 5.

Our unconditional recommendation is to 'climb with the body you've got' and to get to know the strengths and weaknesses of your body. Tall climbers are at a clear advantage when it comes to reachy moves, but will find steep compression climbing more challenging. On the other hand, shorter climbers have to compensate for their lack of reach by being more dynamic in style but will be at an advantage when the climbing is steep and/or features more compression. Heavier climbers will often also have more muscle mass, which will give them an advantage on physically hard climbs, while a lighter climber might find it easier to hang on to smaller holds, as they have a higher finger strength to weight ratio.

**'IF YOU CAN'T DO THE MOVE, THERE IS NOTHING TO
ENDURE.'**

TONI YANIRO



PHOTO: MIKE MULLINS

STRENGTH

As the walls get steeper, the holds smaller and the moves bigger, the need for a strong core, strong arms and strong fingers increases. This is what we call climbing-specific strength.

It's easy to forget how specific training actually is when getting lost in the myriad of training videos, articles, blogs and interviews. For example, if you're doing pull-ups on good holds, you'll get good at doing pull-ups on good holds. And if you're exercising your core by doing the plank, you'll get better at doing the plank. It is an oversimplification to say that general exercise doesn't have an effect, but the direct transfer value to climbing can in many cases be marginal. We must, therefore, focus our strength training on the specific performance factors – finger, arm and upper body strength.



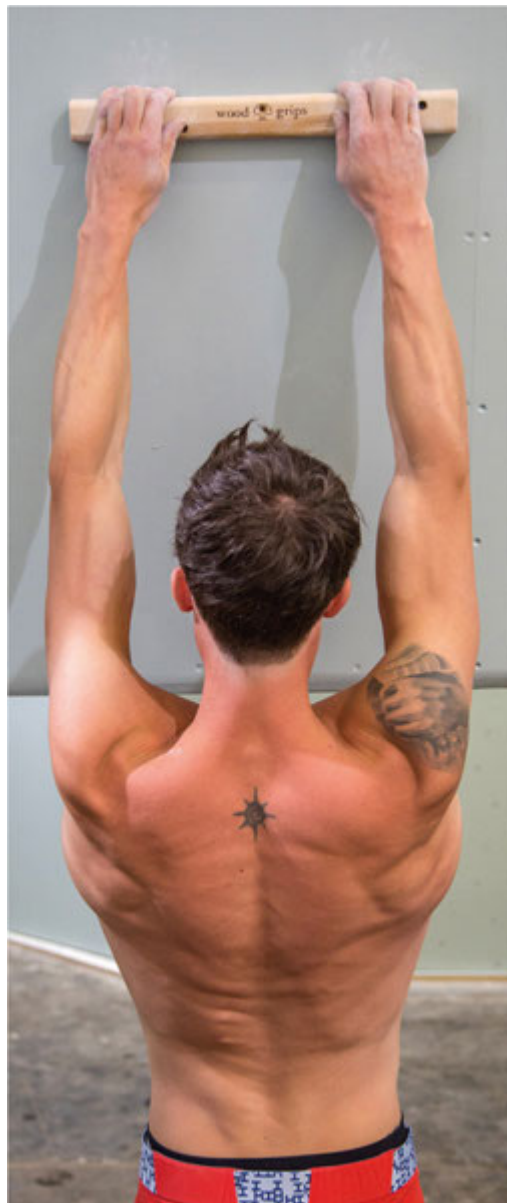
PHOTO: BÅRD LIE HENRIKSEN

Stian Christophersen on the first ascent of Lierpillaren Direkte (F8b+), Bergflødt, Norway.

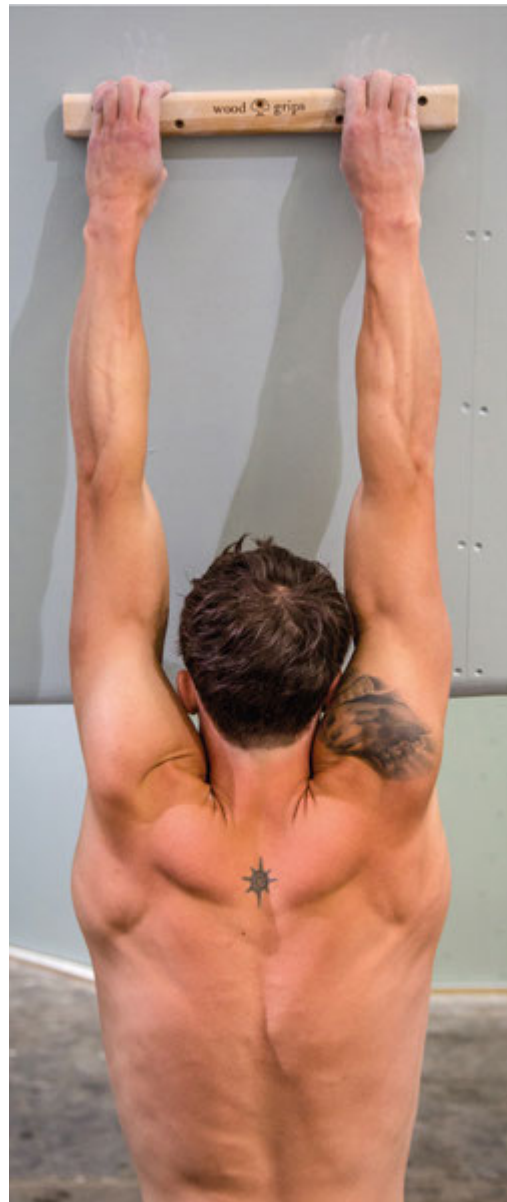
FINGER STRENGTH

Finger strength, as in what is the smallest hold you can still hang on to, is the most important performance factor in climbing. We can compensate quite a bit with good technique, but it's obvious that the best climbers also have the strongest fingers. Finger strength has proven to be a good indicator of what level of climbing you are at and will be a deciding factor for how hard you can pull on a route or boulder problem.

Strong fingers also correlate with good endurance. Put simply, the stronger your fingers are, the poorer the holds you can hold, and the less you'll use of your maximum strength per hold, which all means it will take longer for your forearm muscles to tire. Of course, every time you're climbing, you're also training finger strength, but the best way to get strong fingers specifically is by doing so-called deadhangs.



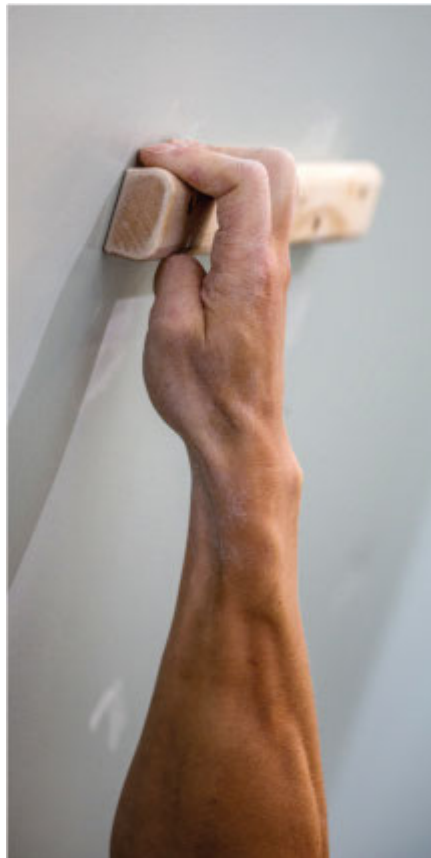
It's important to lower your shoulders and slightly rotate your elbows in towards each other.



Avoid hanging passively with your shoulders up by your ears, as shown in the picture.

DEADHANGS

Finger strength has always been regarded as important, but it is only in recent years that specific finger strength training has gained any real traction in the climbing community in general. Spanish climber and researcher Eva López deserves much of the credit for the increase in knowledge on the subject of finger strength training. Her PhD on the subject brought forth simple and effective training methods by using deadhangs on fingerboards to improve finger strength. The advantage of deadhangs is that the load is easily adjusted based on the level of the climber, and progression is easily tracked. The load is adjusted by increasing/decreasing the size of the hold, by adding/subtracting weight and by adjusting the hang time.



The most useful grip position when training finger strength is the half crimp. Your front three fingers should be bent at 90 degrees at the middle joint, and all four fingers should be engaged on the hold.



If your hand opens, let go. This means the load is too high and you're at risk of injury.

1. SIZE OF THE HOLD

It will be progressively harder to hang a given time interval as the hold becomes shallower. Down to a certain point, around 2 to 4 millimetres, the depth of the hold can be a measure of how strong your fingers are. On shallower holds than this, the anatomy of your fingers and the friction will decide if you can hang or not, and reducing the hold size won't give an increase in muscular training effect.

2. TOTAL WEIGHT

If it's too hard for you to hang with the full weight of your body, you can start by subtracting from your body weight by using an elastic band or a rope pulley system with weights. Stronger, more experienced climbers might have to add extra weight, as they're already accustomed to hanging on small holds with the full load of their body weight.

3. HANG TIME

The smaller the hold you use or the more weight you add, the shorter the time you'll be able to hang on. A shorter hang time with more added weight or a smaller edge size will cause some specific adaptations in muscles, tendons and connective tissues, while longer hang times with less load and a bigger edge size will cause others. It's important to note that you will also have gains in strength with longer hang times as long as you hang on close to failure; strength gains do not only come from short and heavy hangs. To use hang time as a variable, it's easier to vary your training and train with different methods to meet the training goals you've set for yourself.

We recommend using primarily the half crimp grip position. This is the grip position that provides the highest muscular load to the forearms, and it is also of value for other grip positions. But it's important to remember that whatever grip position you're training is also the grip position you'll get strongest at. So, if you want to improve your pulling power on slopers or pockets, then the grip positions you are training should reflect this. It's also important to maintain control of your hang posture. You should pull your shoulders down and back, with your arms straight, and the inside of your elbows facing each other.

KEEP THIS IN MIND!

Deadhanging is specific strength training for your fingers, and the climbing community has traditionally regarded finger-specific training as putting climbers at an increased risk of injury. The recommendation is usually that you should have a certain training base before starting with this method. This base is built through general climbing, and you could also argue that it builds other performance factors rather than just pure finger strength. Working on all factors of your climbing is where your focus should be at the start of your climbing career.





Still, it's important to point out that deadhanging is by no means more stressful for the fingers than hard, fingery bouldering sessions, and well-controlled, specific strength training for the fingers could actually mitigate the risk of finger injuries in the future. It is much easier to regulate the load when doing deadhangs than during a bouldering session, and because strength training, in general, is regarded as one of the single most important factors for reducing the risk of injury across any sport, it's natural to assume that it's possible to achieve a similar effect with specific finger strength training.

Special care must be taken regarding finger-specific training for younger athletes. Because of skeletal development throughout childhood and adolescence, younger athletes may be more susceptible to skeletal injuries following high-intensity finger training methods on fingerboards, campus boards and on the bouldering wall.

DEADHANG TRAINING METHODS

Here are the three methods we recommend for deadhang finger strength training.

1. HANG ON

HANG TIME	REST	REPETITIONS	SETS	BUFFER	GRIP POSITION
30 seconds	3 minutes	1	8	None: hang on until failure*	Vary between half crimp, open hand, pinch and sloper

This is an easy starting point for many climbers. We’re using hang time to failure to set the load, meaning that on the given hold you must be able to hold the grip for over 30 seconds. You can use bigger holds or even a pull-up bar, or take some weight off with a band or a pulley system, to enable you to hold on for the 30 seconds. If you can hang on for a substantially longer period of time (such as 40 seconds or more) you need to use smaller holds or add weight. We find this deadhang method relevant for training all the different grip positions except full crimp. We recommend that you vary between different grip positions during the session.

Start with one session per week and progress to two sessions per week after three to four weeks of training. As you go several times to failure, this session should be done as a standalone session in combination with complementary strength training.



2. STRENGTH HANGS

HANGTIME	REST	REPETITIONS	SETS	REST BETWEEN SETS	GRIP POSITION
10 seconds	5 seconds	4-5	4-5	1 minute	Half crimp

This method is similar to traditional strength training, where each set is close to maximum effort by combining both intensity (hold size and total weight) and volume (number of repetitions). Resting time between sets is short enough not to allow complete recovery between each set, and this helps to fatigue the muscles throughout the session. In practice, this means you should only just complete – or just fail to complete – the last hang of each set, and you should maintain this intensity throughout the whole session.

It should be easy to adjust the size of the hold during the session so that you're able to complete the correct number of reps and sets. If you're unable to hang for the 10-second repetition, it's important that you take some weight off or use bigger holds so that you maintain control for every set. Advanced climbers can also adjust the load by adding extra weight.

Start with one session per week, either as a stand-alone session or after warming up for a regular climbing session. You can do both bouldering or roped climbing after this session, but take into consideration that your forearms and fingers will be fatigued and so you shouldn't overload them during the rest of your climbing session. After three to four weeks you can increase to two sessions per week, with a minimum of two rest days between each deadhang session, and then go on gradually reducing the hold size to maintain a steady training progression. Our advice is to complete one training cycle of the Hang On method before progressing to this method.



3. MAX HANGS

HANGTIME	REST	REPETITIONS	SETS	BUFFER	GRIP POSITION
10 seconds	3 minutes	1	4-5	1-3 seconds	Half crimp

The premise of this exercise is to hang in control for 10 seconds per repetition. Between each hang, you rest for 3 minutes and then repeat for four to five sets. The buffer gives you an indication of how close you are to maximum effort, independent of any reduced or added weight. With a buffer of three seconds, you should be able to hang for a maximum of 13 seconds, but still let go after 10 seconds. The shorter the buffer, the closer you are to maximum effort, but there's still a reasonable training effect with a buffer of up to 3 seconds. This buffer also makes the training a lot safer with regards to injuries as it avoids you just barely being able to hang for 10 seconds or, even worse, falling off uncontrolled.

This method is very well suited to improving your maximum finger strength as the load is high, there are fewer repetitions and the rests are longer. This means you will improve the ability of your muscles and nervous system to develop and sustain force. It's a natural progression to start with one or more cycles of the Hang On method and continue through one or more cycles of Strength Hangs before commencing a cycle of Max Hangs.

As the Max Hang method only involves a few repetitions, and even though the load per hang is high, it allows for the exercise to be combined with a regular climbing session afterwards. We recommend starting with one session per week, then upping to two sessions per week after about four weeks, or when you feel progression in your training. Further progression is easily promoted by adding extra weight.

Deadhanging is a safe and efficient way to train finger strength, but it lacks the aspect of movement which after all is what climbing is all about, and a downside to training deadhangs is that you're only really training to hang. When we're climbing, we're moving between holds, which means we also have to have good arm and upper body strength.



PHOTO: ERICK VIGOUROUX

INTERVIEW WITH

EVA LÓPEZ

Spanish climber and researcher Eva López wrote her PhD on using deadhangs for finger strength training. She compared the effect that two different approaches had on finger strength and endurance.

THESE METHODS WERE named MAW (Maximum Added Weight) and MED (Minimum Edge Depth). The MAW method involved hanging in control for 10 seconds on an 18-millimetre-deep edge with as much added weight as possible, and with a safety buffer of 3 seconds. The hang was repeated four to five times with a 3-minute rest in between each hang, two times per week. The MED method consisted of hanging with no added weight on the smallest hold possible for 10 seconds, plus a 3-second buffer, again repeated four to five times with a 3-minute rest in between each hang, two times per week. The participants in the study warmed up before doing the deadhangs and followed the session with a regular climbing session afterwards.

The training lasted for eight weeks, where one group completed four weeks of MAW followed by four weeks of MED, while the other group did the opposite, namely four weeks of MED followed by four weeks of MAW. The result was that group one (the MAW-MED group) performed better on both strength and endurance tests than group two. Group one had an increase in maximum strength of 10 per cent after four weeks, while group two increased their maximum strength by 2 per cent. This effect did reduce a few weeks after the training period, but even so, this is a significant increase in maximum strength, especially considering that most of the participants were already experienced climbers with an average level of F8a+.

‘AS LONG AS YOU’RE PROGRESSING IN YOUR TRAINING, IT SEEMS TO BE BENEFICIAL TO ALTER BETWEEN PERIODS OF MAW AND MED, WHILE ALSO ADJUSTING THE INTENSITY AND HANG TIME.’



PHOTO: JAVIPEC

Eva López training on the Transgression fingerboard.

When training methods are presented in scientific articles, there are often many questions that arise and nuances that should be highlighted. We asked Eva some questions regarding deadhanging and her research.

Is it important to switch from MAW to MED after four weeks?

First of all, it's important to point out that the training needs to be adjusted to a climber's individual needs, goals and abilities. Every climber – and coach – needs to choose a method, and decide how to incorporate this method into their training plan, based on their training experience, finger strength, goals and so on. This is why I'm reluctant to lay down specific rules, but it's reasonable to say that an advanced-level climber shouldn't use the MAW method before she has completed a moderate-intensity period of MED. For example, she can start with a 15-second hang time with a 5-second buffer for two to three sets the first week, and then increase to four to five sets the fourth week. I've covered this in detail in both my articles and my blog, and I would encourage anyone interested in deadhang training to get acquainted with the material we have gathered.^{1, 2} I also have a YouTube channel where we explain different training methods and hang technique.³

On a general basis, I would always recommend the easiest and least intensive training method and periodisation that still yields an effect. This is an effective, safe and sound way to train. Regarding the specific question, it's logical that after a training period on relatively deep holds with added weight, you'll be ready to train on the smallest holds possible without added weight. As an apropos, hanging on thin edges without added weight is one of the more specific aspects of climbing, and progression in specificity is one of the founding training principles. During our study, we noticed that the increase in finger strength after a period of MAW training continued even after switching to the MED method, even though the strength test was performed on a 15-millimetre edge. Our hypothesis is that the strength increase after four weeks of MAW training enables the climber to use even smaller holds in the following four-week period of MED, which in turn leads to a further increase in finger strength.

As long as you're progressing in your training, it seems to be beneficial to alter between periods of MAW and MED, while also adjusting the intensity and hang time. This is especially the case when starting with deadhang training, and then you can experiment by altering between MAW, MED and other deadhang methods.

On this basis, I would say that going from MAW to MED isn't a general rule, but just one of many strategies for periodisation to choose from. We need to look to the training literature, which states that the stronger and more advanced an athlete is, the higher the training dosage needs to be, and the more we need to vary the training methods, choice of exercises, and periodisation strategies in order to create new stimuli and further development. This way we avoid stagnation and reduced motivation.

‘ON A GENERAL BASIS, I WOULD ALWAYS RECOMMEND THE EASIEST AND LEAST INTENSIVE TRAINING METHOD AND PERIODISATION THAT STILL YIELDS AN EFFECT.’

Why have you chosen to use the half crimp as the grip type?

Are there any added benefits of varying between different grip positions?

I would like to clear up a misunderstanding I experience that many have regarding my work. In experimental research, it's obvious that all variables need to be standardised, and we're therefore forced to choose one out of several possible variables. We chose the half crimp because both we and other researchers, and also the climbers themselves, claim this grip position to be the most common for small holds, and small holds are the most common feature on the hardest sections on a route together with the shape and texture of the hold. This doesn't mean that you only have to train the half crimp. I have never made this claim. The circumstances surrounding an experimental study is one thing, but these are not directly transferable to all types and levels of climbing.

It is a well-known fact that isometric [static] training yields a specific strength effect at the joint angle you're training at, and as long as you have enough time to train, you should train all grip positions if you want to become a well-rounded climber. If you're climbing at an advanced level, at a minimum you should be training both the half crimp and the open hand grip through specific training methods like deadhangs. If you have a lot of time for training, or there are specific grip positions on a project or a crag you wish to train for, you could also use equipment like pinch blocks or hooks for isolated finger training without hanging.

On a general basis, all climbers should train all grip positions through climbing different boulders and routes, and advanced climbers can implement the more specific finger strength training methods in their weekly training. For this group, it will be important to moderate the training volume during the climbing week so they can maintain a high level of quality for the specific exercises.

Are there other finger strength and endurance training methods that are as good or even more effective than the MAW-MED method?

There are, but whether these methods actually give a significant, positive effect will depend on many factors like training experience, periodisation strategies and load control, to mention a few.

With that in mind, we can use training methods and exercises like endurance-oriented deadhangs (intermittent deadhangs), one arm deadhangs (maximum strength training), contrast training (weighted deadhangs followed by explosive deadhangs) and campus boarding, without forgetting climbing on different routes and boulder problems with specific requirements to grip size and wall angles, as well as training with MAW and MED. In addition, we should vary the hang time and intensity for the maximum strength methods and the volume and time for the power endurance methods. Based on this, I recommend alternating between the different parameters and methods every four to eight weeks.



There is a risk involved in having an excessive focus on strengthening the arms and upper body, as this type of training often involves muscular growth and hence an increase in body weight. We should try to train with high load and relatively few repetitions to maximise strength while at the same time limiting muscle growth. Implementing this training into more general climbing training requires planning so that the fingers are able to adapt to the increased load from stronger arms and upper body, and potentially a heavier body as a result of increased muscle mass.

ARM AND UPPER BODY STRENGTH

It is well documented that the physical demands on the arms and upper body increase as the climbing becomes steeper and the moves bigger. Still, there are examples of climbers with relatively poor arm strength who perform exceptionally well on steep climbs. This highlights the complexity of climbing, where you can compensate for deficiency in one area by being good in another; but these climbers could perhaps raise their level even higher by training their arm and upper body strength.

At lower levels, bouldering and steep routes can provide enough strength training for the arms and upper body, and through this lay a good foundation for the more specific training to come later. Wall angles are a useful indicator of load, as the climbing becomes more physically challenging as the walls get steeper. A clear advantage of using climbing as a training method is, as previously mentioned, that you also train all the other factors that impact your performance.

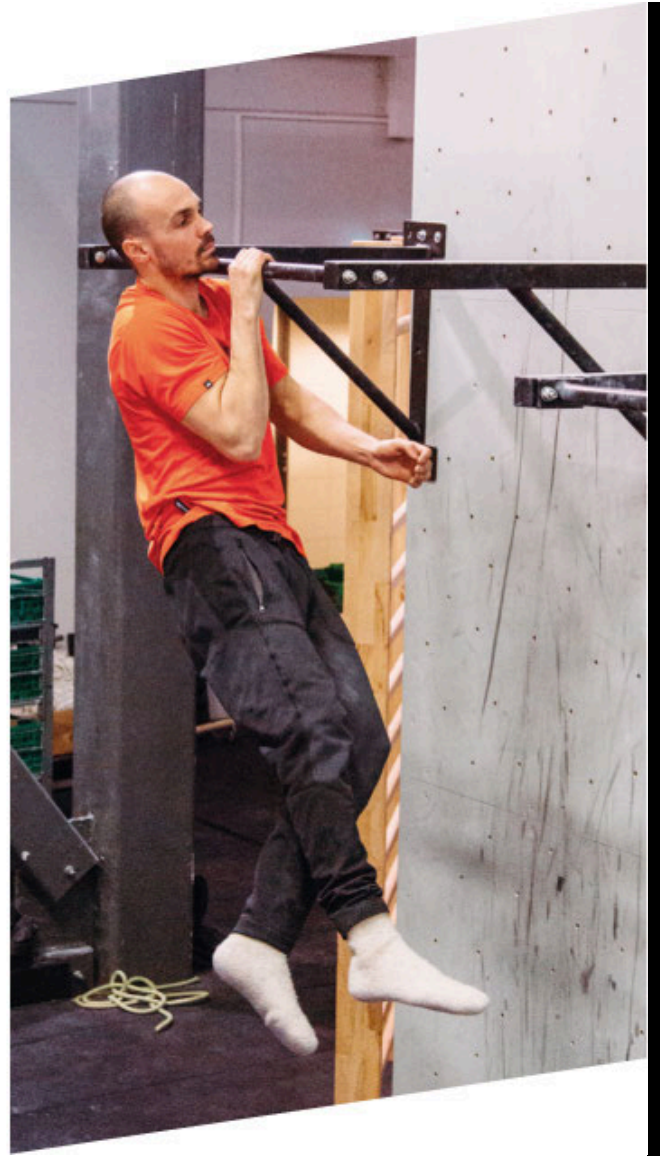
Still, we recommend you add some specific strength training on top of what you would otherwise get through just regular climbing. Variations on pull-ups and lock-offs are the best options to add.



PULL-UPS

Pull-ups are the most common and simplest way to train your pulling apparatus, and for many, this is the first exercise to add in a specific strength training programme. The movement starts with the arms straight, then you pull your shoulders down, rotate your elbows toward each other and bend your elbows all the way until your chin is level with your hands. Try to do the pull-up calmly and smoothly, and don't cheat by 'kipping' and using your legs. If you find doing a pull-up is too hard, then you can use a bungee (as in the photos) or a partner to help you take some of the load off. To begin with, you can try to do as many pull-ups as you can in one go, rest for 2 to 3 minutes, and repeat for three to four sets. When you're able to do more than six repetitions per set, it's time to up the difficulty, either by adding extra weight or by using an offset grip position – meaning one hand is lower than the other. Adjust the difficulty so you're only able to do three to four reps per set, and work your way back up six reps before upping the difficulty yet again. We recommend varying your pull-up training by alternating between added weight and offset grips; this mimics the varied movement patterns of climbing.

Strong, advanced climbers can also try doing one-arm pull-ups. When transitioning from regular pull-ups to one-arm pull-ups, it can be helpful to adjust the load with a bungee cord. This enables the movement to be performed with sufficient quality and allows a gradual progression in the training.



LOCK-OFFS

It's important to be able to lock your arm and body off in different positions, for example when doing a static move or when clipping a quickdraw. Specific lock-off training can be done with one or both arms, depending on your level. Beginners can train by doing so-called 'Frenchies' – locking off during a pull-up at three different stages, for example when your elbow is bent at 45, 90 and 120 degrees. Advanced climbers can do the same exercise with just one arm, and hold each position for 3 to 5 seconds. When transitioning from being able to do regular Frenchies on both arms to doing them with just one arm, try using a bungee cord or get help from a partner. It's important to be aware that training static positions will only make you stronger at holding that exact position, plus or minus a few degrees, so it's important to vary the angle of your elbows to build static strength throughout the range.



COMBINING FINGER, ARM AND UPPER BODY STRENGTH

Up until this point we have looked at how we can specifically train our fingers, arms and the upper body, and we will now look at how we can combine these elements in our training. This kind of combination training is important to make our strength training specific enough to the movement tempo and pattern we face when we're climbing.

We must always position our body in order to maximise the use of the holds. This means we have to create force in different directions as we move between the holds, and that we have to use different grip positions depending on how the holds are shaped. Hanging on edges and training pull-ups and lock-offs is good strength training but it excludes the wide variety of movement patterns we face when climbing.

The strength of a muscle is not the same as how fast it can generate force. What makes hard strength training challenging for your arms and upper body is the fact that the movements are rather slow. And even though it's important to have the strength to maintain a body position or climb slowly, climbers, and especially boulderers, need to be able to generate power quickly, an ability best developed through explosive strength training. In short, the way to train this is by doing these moves as quickly as possible. We differentiate between strength and explosiveness in the fingers in the same way that we do with the arms and the upper body. We need finger strength so we can hang on to holds, but we also need the muscles in the forearm to generate power quickly enough for us to achieve maximum power on a hold in a very short amount of time. This ability is known as contact strength.

By combining campusing (climbing without feet) and campus boarding, we will, in addition to increasing the speed of our arm movements, challenge the contact strength of our fingers even more so than by doing deadhang training. Bouldering will give us a large variation in the types of moves that we do. Through this, we train both slowly generating maximum power when moving statically for a bad hold or maintaining a body position, and also rapid generation of power when moving fast.

BOULDERING

Our experience is that bouldering on progressively smaller holds and steeper walls provides a very good training effect for finger, arm, and upper body strength, and should be considered the most important training method for climbers at any level. We could, for example, mention the fact that both of us were Font 8b boulderers and F8c route climbers before starting with systematic finger strength training. To get the most out of a bouldering session they have to be planned in advance to ensure you're training the specific abilities that you want to develop. Most importantly, plan which type of boulder problems you want to work with, and make sure to get enough rest in between each attempt so that it's actually a strength training session and not about endurance.

It'll take some experimentation to find the right problems for a specific session, so we recommend that you use some sessions to put together a good selection of different boulder problems. Physically, you'll get more out of the training if you remove the technical tricks you can use to take some of the load off of your fingers – like heel and toe hooks and twisting in. On a general basis, we consider the physical effect on your fingers, arms and upper body to be higher if you front the wall and use small footholds.



1. MAXIMUM INTENSITY BOULDERING

Choose four to five boulder problems close to your maximum level, with four to five moves per problem. Repeat each problem two to three times with at least 3 minutes of rest in between each attempt. If the main goal of the session is targeting explosiveness, each boulder problem should consist mainly of large moves that require rapid development of force both to start and to stop the movement. Alternatively, when targeting finger, arm and upper body strength, it's usually sufficient that the wall is steep and the holds are relatively small, without placing too much emphasis on the distance between the holds.

2. PYRAMIDS

An alternative to the session described above is to build the difficulty of the problems like a pyramid, which will have you doing more boulder problems. This is a common way to structure a session, but it's important to know that it's the hardest problem that will have the greatest effect on your strength. As above, you should decide if you want to challenge your explosiveness or your strength.

A pyramids bouldering session might look something like this:

6a	6b	6c	6b	6a
3 problems	2	1 (two tries)	2	3

Choose different types of boulder problems, and don't have more than two attempts on each problem. Rest for at least 3 minutes between each attempt. Adjust the difficulty of the boulder problems to your individual level. We recommend that the hardest problem is one to two grades below your maximum. So, if your personal best is a Font 7b, make the hardest problem in the pyramid a 7a or 7a+.





3. MOONBOARD

Developed by climbing legend Ben Moon, the Moonboard is perhaps the most commonly used system training wall around today. It is 40 degrees overhanging and features boulder problems set using a systemised grid of small, positive holds. You'll find suggestions for problems in the Moonboard app, which lets you climb on thousands of predefined problems at all levels of difficulty. A bouldering session on the Moonboard is very good training for your finger, arm and upper body strength, and it lets you do both maximum intensity sessions and pyramid training.



GOING FOOTLESS

Footless climbing, or campusing, involves doing climbing moves without using your feet. This is great training for your fingers, arms and upper body, as you're not using your legs to help. To begin with, you should focus on landing the holds with your arms bent and maintaining control over your body to avoid taking any unnecessary swings on straight arms – so start your campus training with good holds and easy moves. Start by doing three to four moves per problem, and increase to six to eight moves as you progress. A positive aspect of campusing is that you can vary the moves and vary between doing them slowly for strength or fast for explosiveness. When training explosiveness, we recommend doing fewer, but bigger moves; ideally no more than two to three moves per problem. Advanced climbers can reduce the hold size for an even greater finger strength training effect.

THE CAMPUS BOARD

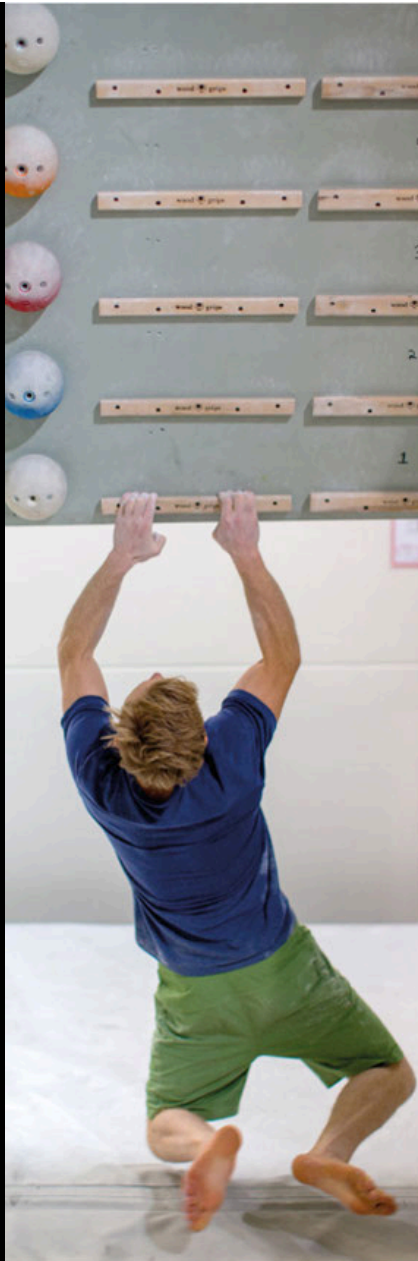
The campus board is one of the earliest forms of training installations developed especially for climbing. It was designed by Wolfgang Güllich as part of his systematic training for the first ascent of Action Direct – the first 9a route in the world – in the Frankenjura, Germany, in 1991. The name of the board comes from the college gym it was first installed in – the Campus Gym. The campus board consists of wooden edges in different sizes, mounted on an overhanging wooden board, and allows for different methods of training finger and arm strength. The advantage of campus boarding when compared to deadhangs is that it involves arm movement – meaning you go from one edge to another and therefore let go of one hold to grab another, just like regular climbing. This results in a very specific training effect for fingers, arms and upper body, in addition to training technical elements like coordination and dynamics. The challenge with campus boarding is the high risk of injury to fingers, elbows and shoulders. Younger athletes should postpone campus board training until their bodies are fully developed so as to avoid risking injuries to the skeletal structure in their fingers.



Campus training should be done at the beginning of your training session so that you're well rested and can maintain quality in your training.

Fully developed climbers can stay safe and injury free on the campus board by using proper technique and a sensible progression. First and foremost, it's important to learn how to properly move on the campus board, and for many, the best way to do this is by practising the moves while keeping the feet on footholds placed below the board. Start by doing shorter moves on the deepest edges, progress by skipping holds and making bigger moves, and then move on to smaller holds as your technique improves.





1. TRADITIONAL CAMPUS BOARDING

It's important to maintain control of the movement so that you land each edge with four fingers in a half crimp grip position and avoid landing on a straight arm. To achieve this, these exercises are targeted towards strength training rather than explosiveness, and will therefore build a foundation for the more explosive campus training.

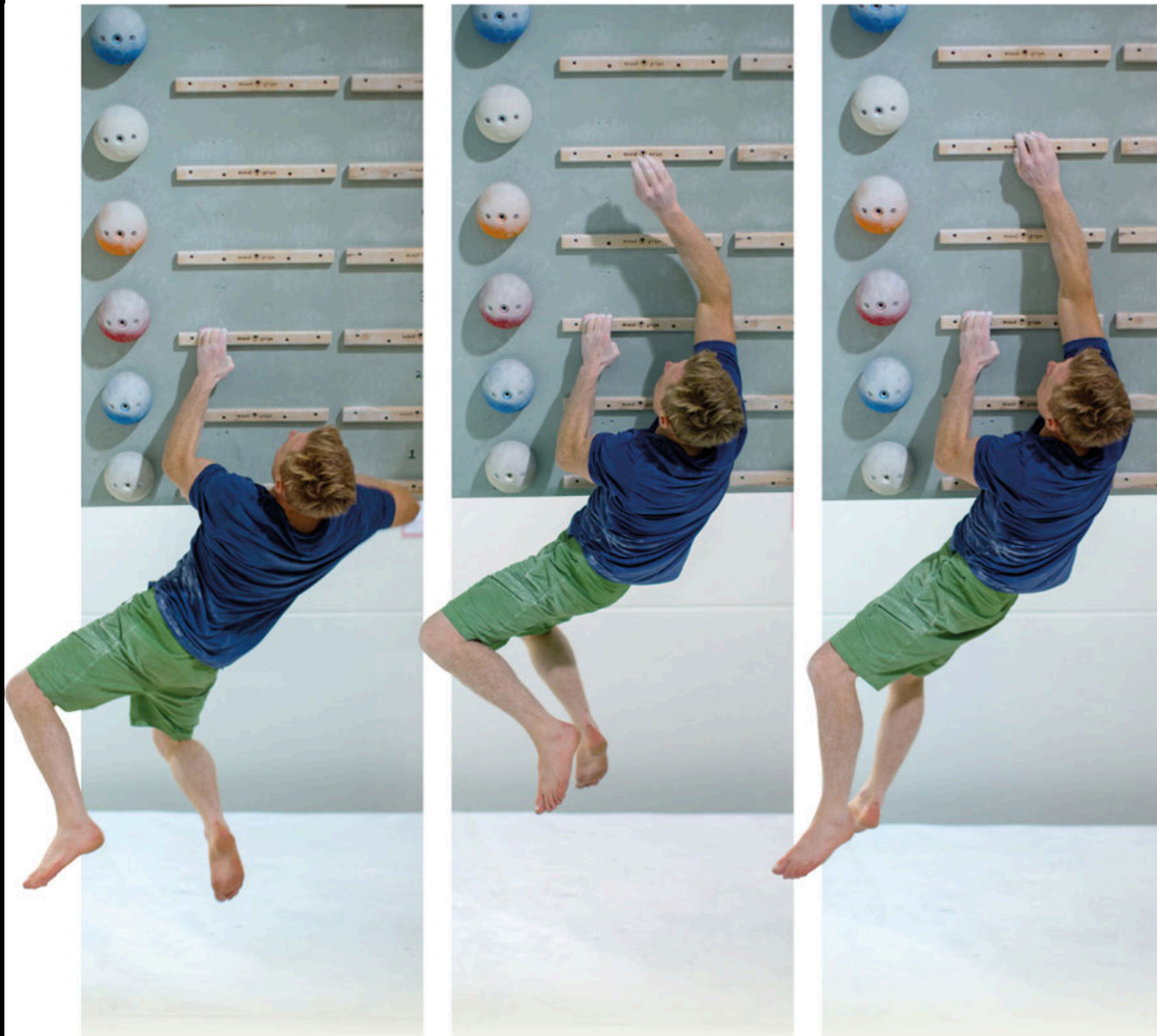
1. Up-down (1-2-1, alternating hands)

2. 1-2 match, 2-1 match

3. 1-2-3 match, 3-2-1 match

4. 1-2-4 match

5. 1-3-4 match



When you feel ready to increase the difficulty, you can do the same exercises but without your feet on the wall below. When you're comfortable going footless, increase the distance between the edges, and then choose smaller edges. Here are some examples to try:

1. 1-3-5

2. 1-3-6

3. 1-4-7

4. 1-5-8

EXAMPLE OF A CAMPUS BOARD SESSION

Warm up by doing some exercises with your feet on the wall.

First set:

1-3-5 x3 per arm, with 1 minute of rest between each repetition.

Rest for 3 minutes.

Second set:

1-3-6 x2 per arm, with 1 minute of rest between each repetition.

No rest.

Third set:

1-2-3-4-5-6-7 x3 with 1 minute of rest between each repetition.



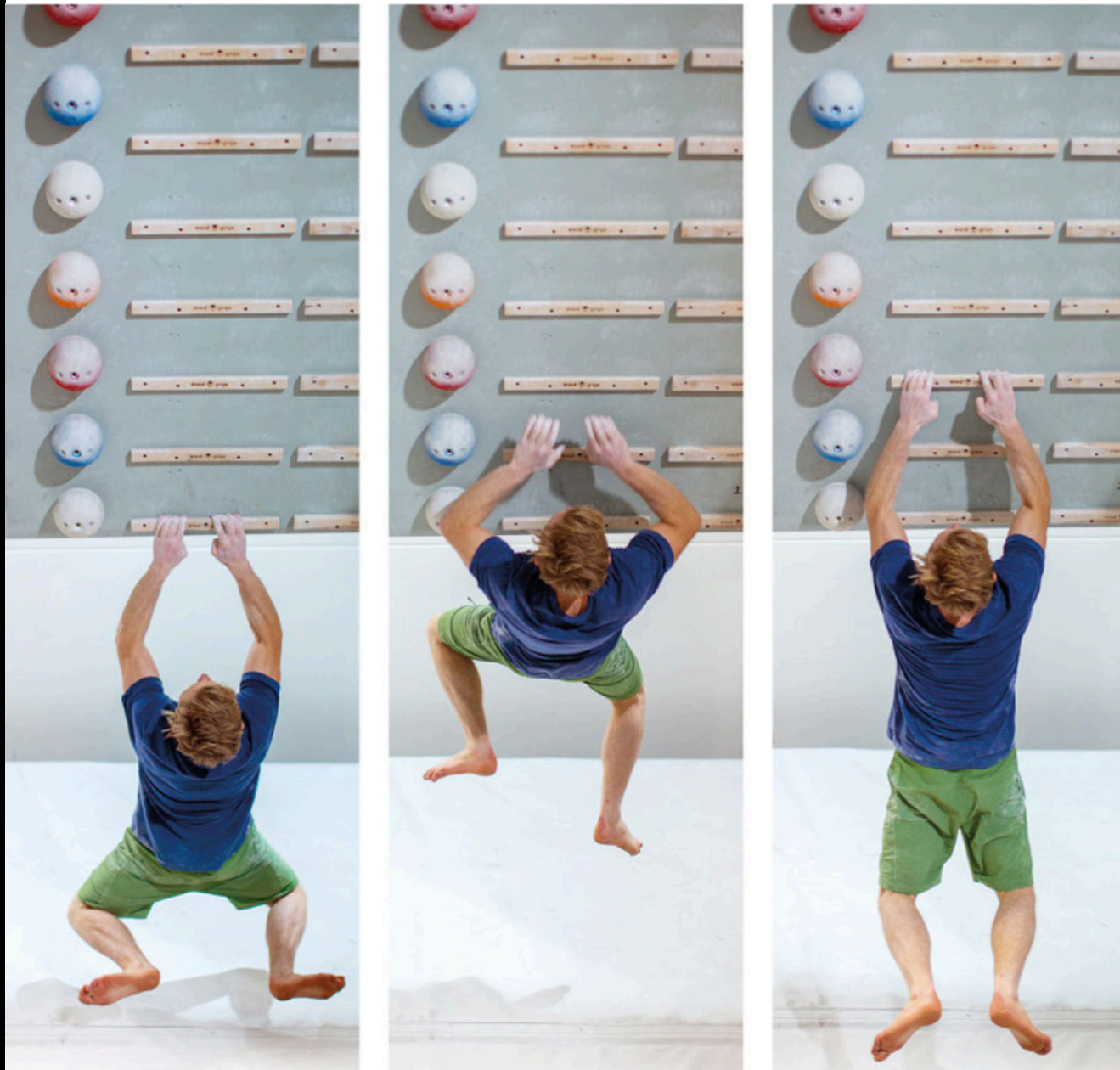
2. EXPLOSIVE CAMPUS BOARDING

When doing campus moves to train explosive strength and finger contact strength, each move should be as big as possible. Again, we must point out that this is not a suitable training method for younger climbers because of the risk of injury. Exercises we recommend are:

1. 1 to maximum height with one hand
2. 1 to maximum height with both hands
3. 1-2-3-4-5-6 maximum speed one hand at a time

4. 1-3-2-4-3-5 dyno both hands at the same time (double dyno)

5. 3-1-4 double dyno



Double dynoing on the campus board is one of the hardest and most challenging moves you can do, and when stringing together several double dynos in a row you have to slow your body down before starting up again every time. This form of training can be very effective but it is also closely related to the risk of injury. We recommend that double dynoing is reserved for advanced climbers with an established training base and experience with systematic, traditional campus board training.

EXAMPLE OF A CAMPUS BOARD SESSION

Start with exercise 1:

Three repetitions per arm, with a 2-minute rest between each repetition.

Then do exercise 2:

Four repetitions, with a 2-minute rest between each repetition.

Finish with exercise 3:

Two repetitions, with a 2-minute rest between each repetition.



CORE TRAINING

Several of the exercises described previously will train your stomach and back – what we call the core muscles – while in parallel also training your arms and upper body. But still, it can do you good to also focus more specifically on your core muscles, as a strong stomach and back is important for several reasons: to maintain body position when moving an arm or a leg; to maintain pressure on footholds during steep climbing; and to stop a swing when your feet do slip during steep climbing.

The most important thing to understand about training your core is that – as we've mentioned previously – training is specific. This means that the training you do has to resemble as closely as possible the movement patterns of the climbing you'll be doing, and this is where your average core workout won't give the desired effect. An obvious example of this is hanging exercises vs floor exercises: the former has the potential to be climbing specific, but the latter does not. In addition, hanging exercises have the added benefit of training your arms and upper body, which allows you to combine multiple elements in your training.

Common for all the following exercises is that you're moving your legs and body from a hanging position. To increase the difficulty of an exercise, you can make the holds worse or try to move your feet further away from your body.

For most of us, a session will be limited by how tired we get in the fingers and forearms while still having plenty left in the bigger muscle groups. Adding two or three of the following exercises to your climbing sessions, two to three times per week, can be an effective way of maximising the time you have available for training, and increase the training yield per workout. Vary between different exercises for each week.

FOOT PLACEMENT

Being able to place your feet on holds from a hanging position is important on steep walls. It is also important that you are able to maintain your body position when your feet are finally back on the wall. Here are two specific exercises to train this ability:

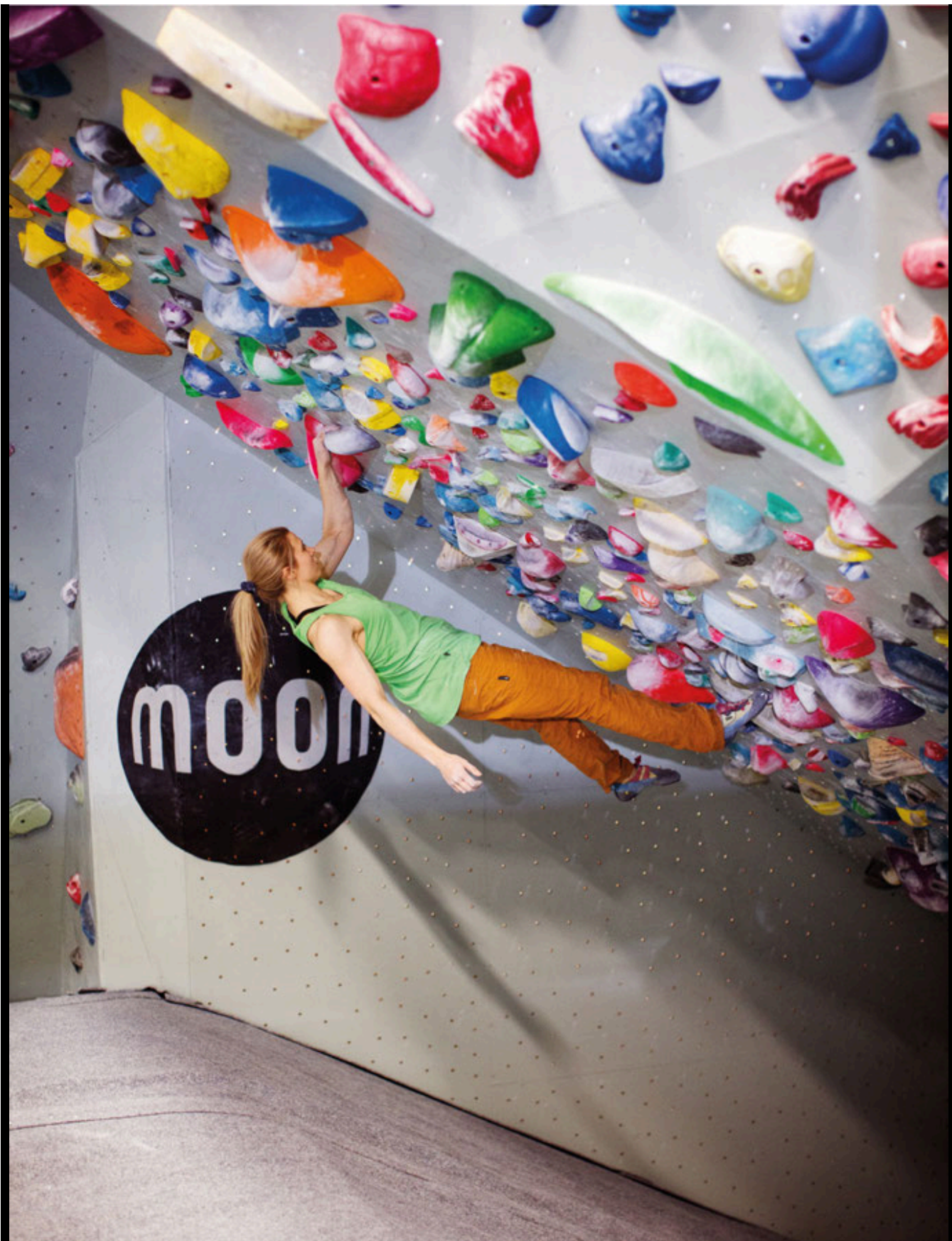


1. TIC-TAC

Choose two good holds that enable you to hang for 30 seconds without using your feet. Choose three footholds on each side of your fall line – one low, one knee high and one at hip height. Place your right foot on the bottom right hold, let go and drop your foot back down, place it on the middle right hold, drop it back down, and then go for the top right hold. Repeat this with your left foot before taking a two-minute rest, and then repeat the full set two to four times.

As an alternative to Tic-Tac, you can hang in the same position but cross your legs over to the other side, so your right foot goes for holds on your left side, and vice versa.





2. DIAGONAL

Hang from one good hold with both hands and place your right foot on a hold far down to your right. Let go with your right hand and hold this position for about five seconds. Repeat on your left side and do three to four repetitions per side, and two to four sets.

Progression for both of these exercises is controlled by the size of the handholds and how steep the wall is. Less experienced climbers can start by using good holds on a 30-degree overhanging wall and move on to a steeper wall before using poorer holds. You can also adjust the difficulty by moving the footholds further away and making them smaller.

HANGING EXERCISES

The following exercises are performed from a hanging position. To ensure the focus is on core strength it's best if they are done using really good handholds, so that finger strength isn't the limiting factor.



1. L-SIT

Start this exercise by pulling yourself up slightly, as when initiating a pull-up, and lifting your legs so that your body looks like the letter L. Many of you might struggle a bit with hanging with your elbows slightly bent. In this case, you can use a bungee to take some of the load off your arms so you're able to hold the position.





To make it even easier, if you're unable to keep your legs pointing straight out, you can bend your knees, or get help from a partner to lift your legs and then try to hold back as they drift back down.

Alternatively, you can hang with your arms completely straight to further isolate the lower part of your stomach. For climbers who are comfortable with the L-sit, this can be a good way to vary the exercise. For further progression with the L-sit, try swinging one leg out to the side while keeping the other leg pointing straight ahead, then switching back and forth from side to side.

We recommend doing eight to ten repetitions for three sets where each set should last until exhaustion, with 3 minutes of rest between each set.



2. ROOF TOUCH

The roof touch is similar to the L-sit, but here your arms should be straight as you lift your feet all the way up until they're level with your hands. When you're comfortable doing the L-sit without any help and you have energy to spare after each set it's time to progress to the roof touch. You can add variation to the roof touch by alternating between lifting your legs straight up and diagonally – meaning your right leg should be on top when rotating to the left and vice versa. This exercise is significantly harder than the L-sit, and we recommend three sets of four to six repetitions, with 3 minutes of rest between each set.



3. WINDSCREEN WIPERS

For this exercise, start by doing a roof touch, and then lower your legs by rotating sideways until they are horizontal. Continue by rotating back and forth to both sides. As with the roof touch, this is a challenging exercise, and we recommend three sets of four to six repetitions, with a 3-minute rest between each set.

FRONT LEVER

This is perhaps the most well-known strength exercise for climbing and it's definitely one of the hardest. To do a perfect front lever your arms and legs should be completely straight and your body horizontal. To get to this level we can divide the progression ladder into multiple steps:



1. ASSISTED FRONT LEVER

Lift your legs up and get into the front lever position. Use a bungee or a partner to help you keep your hips up and hold this position for 3 to 5 seconds. Repeat for three to four sets.

Getting assistance like this is of course also an option for the one-legged front lever.



2. ONE-LEGGED FRONT LEVER

Lean back and lift your hips up towards the roof. Extend one leg forwards horizontally, but tuck your other leg up towards your chest to reduce the load. Hold this position for 3 to 5 seconds before dropping down. Repeat three to four times for each leg.



Do the exercises with your arms slightly bent to start with, and straighten them as you progress.



3. FRONT LEVER

You're now hanging horizontally, and both your arms and your body should be straight. If you've got this far, try working on holding the position for longer, or even adding extra weight around your feet to make it harder still.

The most challenging part of the front lever is stopping your hips from dropping. The goal of this exercise is to keep your hips level with your shoulders and your body horizontal. There's nothing wrong with training for the full front lever if you can't yet manage the one-legged front lever without your hips dropping, but keeping your hips up is where your focus should lie. Using a bungee or a training partner can be very helpful to ensure you're doing the exercise correctly.